POWER PULSE

Protective Relays Important Formulae

Notes by Balaji Sir

1 Overcurrent relay

1. $$PSM = \frac{I_{fault}\ (\text{CT secondary})}{\text{plug setting} \times I_{rated,relay}} = \frac{I_f}{CTR \times PS \times I_n}$$

2. $$t = TMS \times \frac{k}{PSM^{\alpha} - 1}: \quad \text{SI: } \frac{0.14}{PSM^{0.02} - 1},\ \text{VI: } \frac{13.5}{PSM - 1},\ \text{EI: } \frac{80}{PSM^2 - 1},\ \text{LTI: } \frac{120}{PSM - 1}$$

IEC standard inverse curves

3. $$\text{Grading margin} = t_{CB} + t_{overshoot} + t_{safety} \approx 0.4\ \text{s}$$

2 Distance relay

4. $$Z_{sec} = Z_{pri} \times \frac{CTR}{VTR}, \qquad CTR = \frac{I_p}{I_s},\ VTR = \frac{V_p}{V_s}$$

relay settings in secondary ohms

5. $$Z_1 \approx 80\%,\ \ Z_2 \approx 120\%\ (\text{or } 100\% + 50\%\ \text{of next}),\ \ Z_3 \approx \text{backup}$$

zone reaches (of the protected line)

6. $$\text{Operates if } Z_{apparent} = \frac{V}{I} < Z_{set}$$

3 Torque equations

7. $$T = K_1 I^2 - K_2 V^2 - K_3\ \ (\text{impedance: operates for } \frac{V}{I} < \sqrt{K_1/K_2})$$

8. $$T = K\,V I \cos(\theta - \tau) - K_0\ \ (\text{directional})$$

9. $$T = K_1 I^2 + K_2 V^2 + K_3 V I \cos(\theta - \tau) + K\ \ (\text{universal relay torque})$$

10. $$\text{Reset ratio} = \frac{\text{drop-out value}}{\text{pick-up value}} < 1$$

reset ratio 0.9 means drop-out at 90% of pick-up

11. $$\text{Earth fault setting} = \frac{I_{earth}}{I_{full\ load}} \times 100\ \%\ \ (\text{typically } 10\text{–}40\%)$$